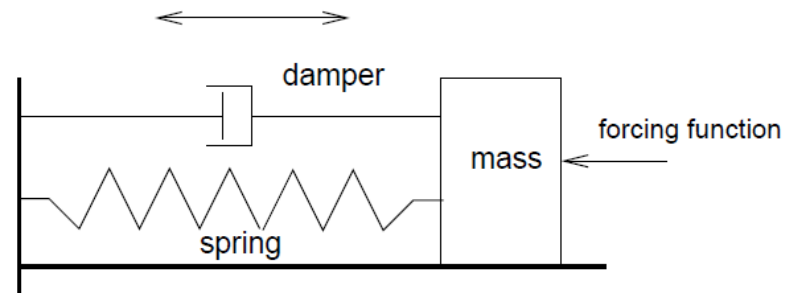


The physics of sounds

Resonant systems

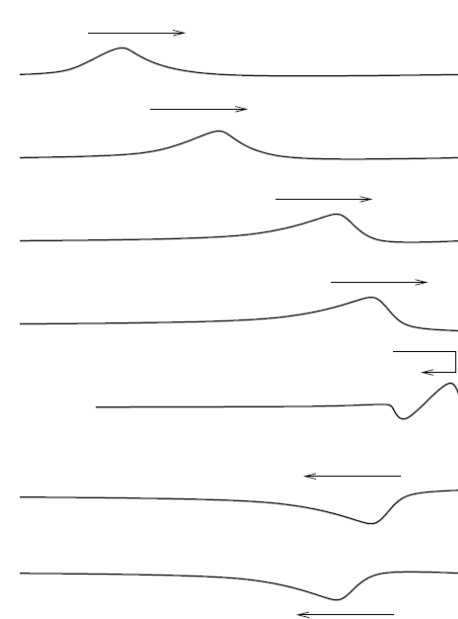
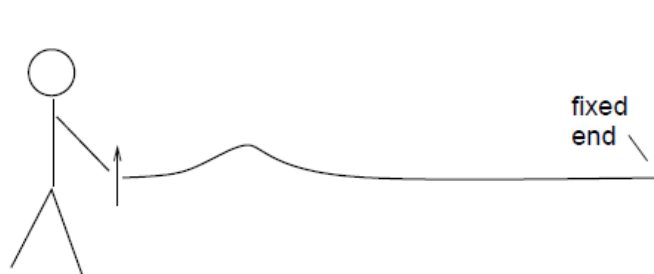
- Consider the mass spring shown below
 - If you displace the mass, the system will try to return to its rest position
 - In the process, it will lead to oscillations around the rest position
 - Due to frictions, the mass will eventually settle onto the rest position
- Now consider a periodic forcing function being applied
 - At a certain frequency f_R , the size of the oscillations will increase over time rather than decrease
 - Eventually, and in the absence of other factors, the system will break
 - Frequency f_R is known as the resonant frequency of the system

$$f_R = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

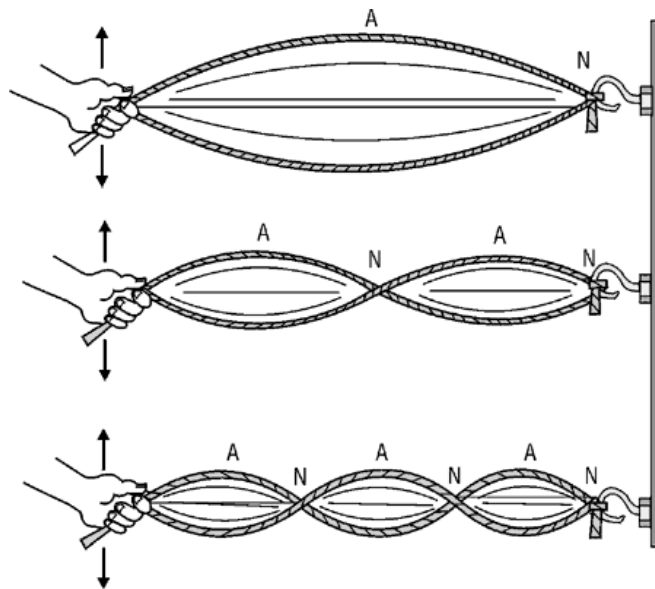


Travelling and standing waves

- Consider a person holding a rope fixed to a wall on the opposite end
 - The person gives a jerk to the rope, and as a result a wave forms and starts travelling down the rope
 - When the rope reaches the wall, it is reflected and begins to travel back towards the person
 - When it reaches the person, the wave is reflected back towards the wall
 - This process goes on until all energy in the rope dissipates



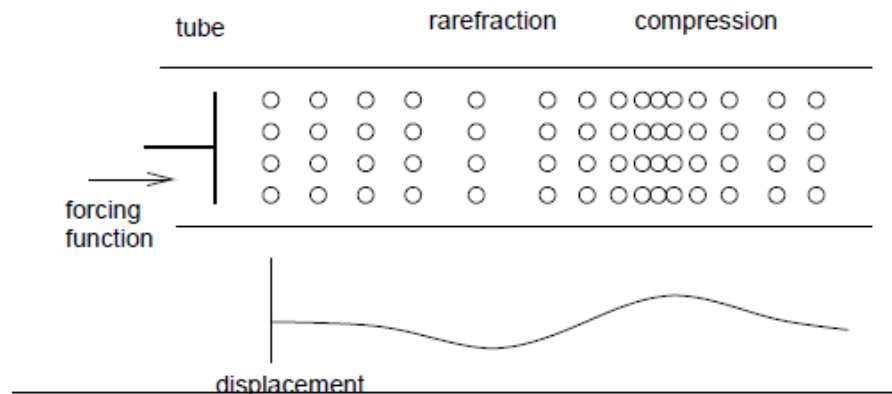
- Now consider the person performs a repetitive movement
 - The forward and backward wave will interfere, which may lead to full cancellation (if the two waves are in anti-phase)
 - At some frequencies, the reflected wave will reinforce the forward wave, and the rope settles into a fixed pattern
 - The resulting wave will appear not to be moving at all (a standing wave)
 - Thus the rope acts as a resonator: it amplifies some waves and attenuates others



- What determines the behavior of the system?
 - The frequency of the oscillations is determined entirely by the hand
 - The rate of travel of the wave is determined entirely by the rope
 - Boundary conditions: whether the rope is fixed or free at each end.
- What is the relationship with the speech signal?
 - The hand acts as the source (the glottal pulses)
 - The rope acts as the filter (the vocal tract)

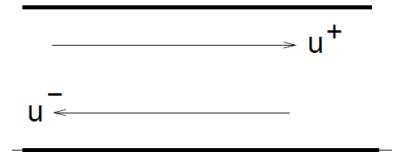
Acoustic waves

- Properties of sound waves traveling in a tube (i.e., vocal tract) are similar to those moving down a rope
 - The effect of a sound source causes air particles to move back and forth, so the wave spreads from the source
 - In some areas, particles come close together (compression) whereas in others they move further apart (rarefaction).
 - One difference is that sound waves are longitudinal whereas those in a rope are transverse, but otherwise the same mathematical model can be used for both systems



Acoustic reflection and tube models

- As with the rope, boundary conditions in the tube will determine how acoustic waves are reflected at the end of the tube
 - At certain frequencies, determined by the length of the tube and the speed of sound, the backward and forward waves will reinforce each other and cause resonances
- We can model the volume velocity (e.g., particle velocity $\times$ area) at position x and time t as:



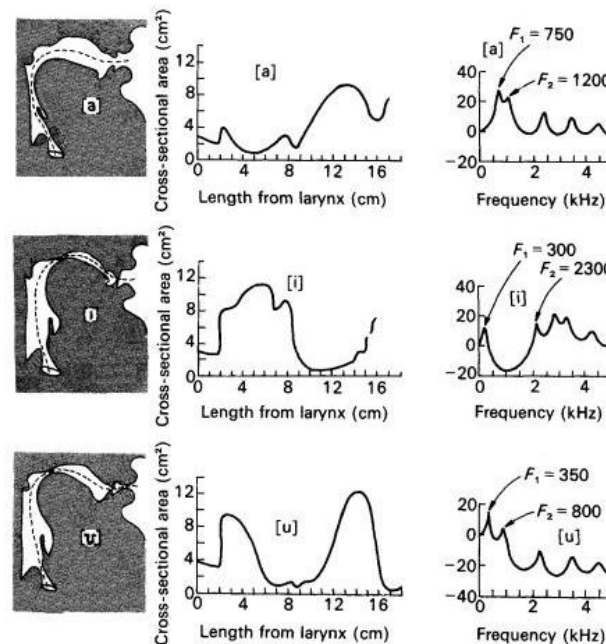
$$u(x, t) = u^+(t - x/c) - u^-(t + x/c)$$

- where $u^+(t)$ and $u^-(t)$ are the forward- and backward-travelling waves, and c is the speed of sound
- And the pressure becomes

$$p(x, t) = \frac{\rho c}{A} (u^+(t - x/c) + u^-(t + x/c))$$

- where $\rho c/A$ is the characteristic impedance of the tube
- Notice how in this case the two waves add up as they meet at point x

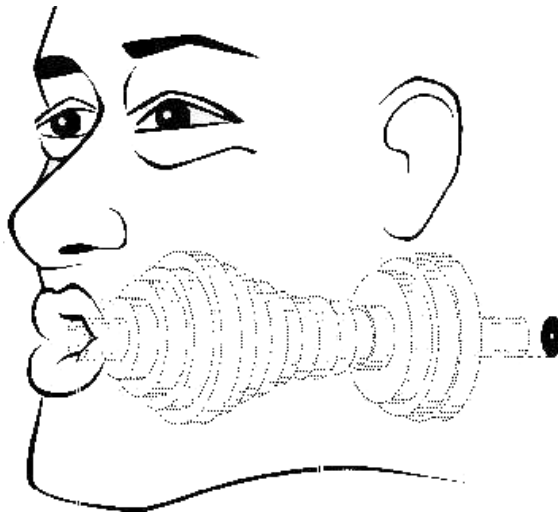
- If the area of the tube remains constant, the wave propagates through the tube
- However, if the area changes, then the impedance changes, which causes reflection, which in turn leads to standing waves, which then cause resonances
- Thus, the impedance pattern on the tube determines the resonance properties of the model



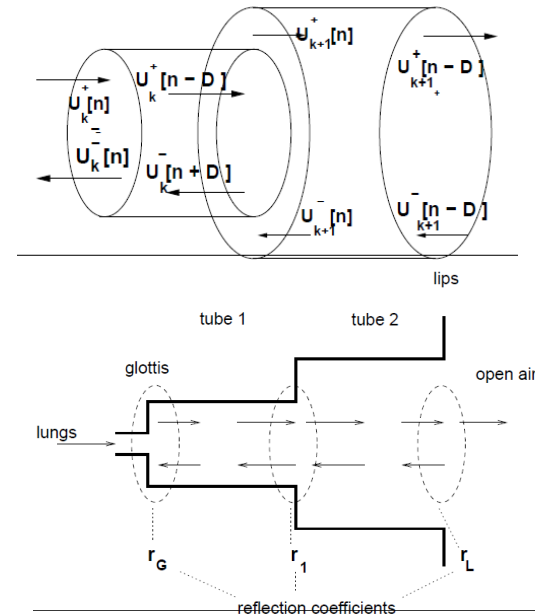
http://www.livingcontrolsyste.ms.com/festschrift/nevin_files/image015.jpg

- The vocal tract can then be modeled as a series of short uniform tubes connected in series
 - By increasing the number of tubes, the vocal tract can be modeled with arbitrary accuracy

As we will see in linear predictive coding , a tube model with N • sections leads to $N/2$ resonances, so in practice only a few tube sections are needed to model the main formants in the speech signal



http://www.gregandmel.net/burnett_thesis/2_3.ht7.jpg



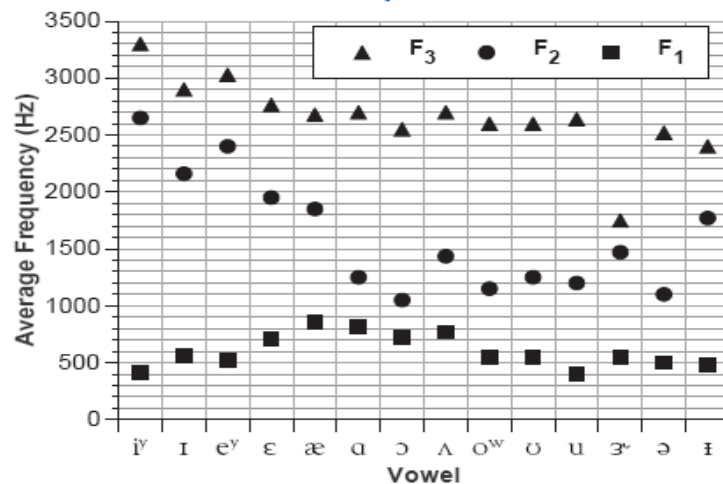
English Phonemes

<i>PHONEME</i>	<i>EXAMPLE</i>	<i>PHONEME</i>	<i>EXAMPLE</i>	<i>PHONEME</i>	<i>EXAMPLE</i>
/i ^y /	beat	/s/	see	/w/	wet
/ɪ/	bit	/ʃ/	she	/r/	red
/e ^y /	bait	/f/	fee	/l/	let
/ɛ/	bet	/θ/	thief	/y/	yet
/æ/	bat	/z/	z	/m/	meet
/ɑ/	Bob	/ʒ/	Gigi	/n/	neat
/ɔ/	bought	/v/	v	/ŋ/	sing
/ʌ/	but	/ð/	thee	/č/	church
/o ^w /	boat	/p/	pea	/j/	judge
/ʊ/	book	/t/	tea	/h/	heat
/u ^w /	boot	/k/	key		
/ɜ ^v /	Burt	/b/	bee		
/ɑ ^y /	bite	/d/	Dee		
/ɔ ^y /	Boyd	/g/	geese		
/ɑ ^w /	bout				
/ə/	about				

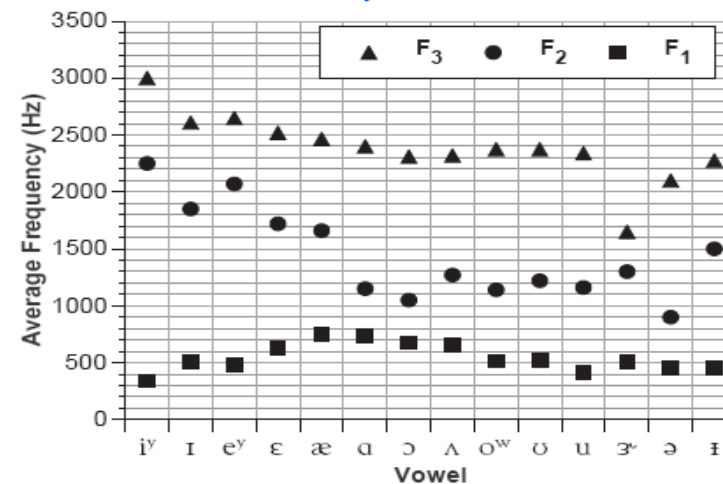
Formants

- Vowels are often characterized by the lower three formants
- **High/Low** is correlated with the first formant, F1
- **Front/Back** is correlated with the second formant, F2
- **Retroflexion** is marked by a low third formant, F3

Female Speakers



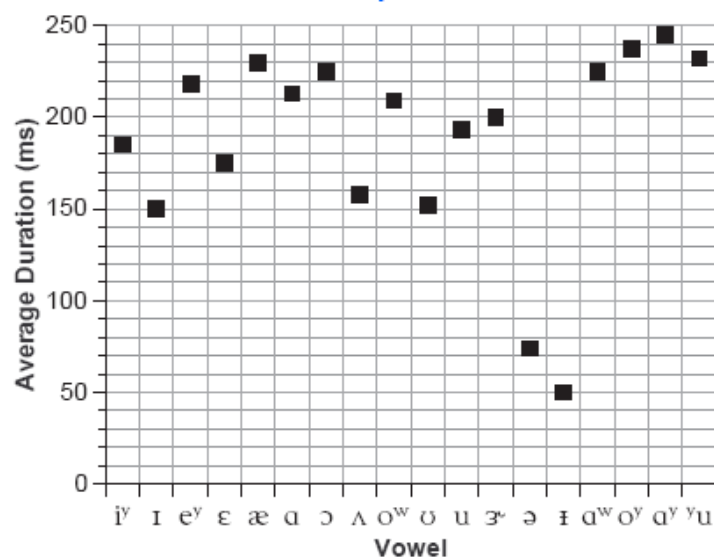
Male Speakers



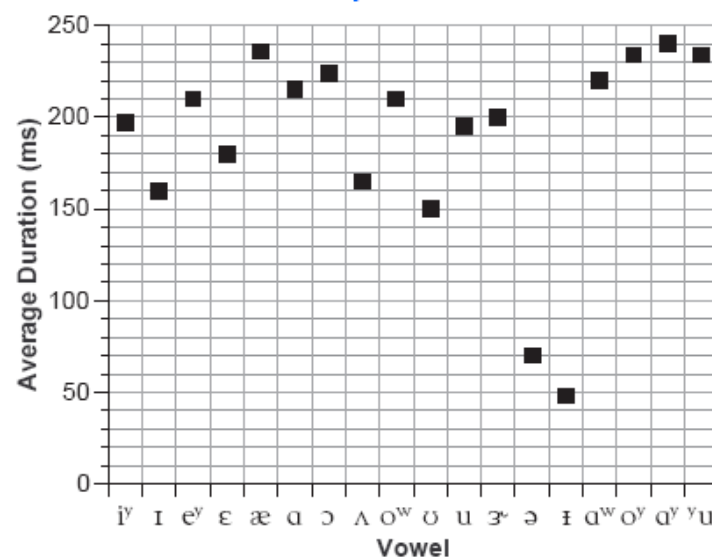
Vowel duration

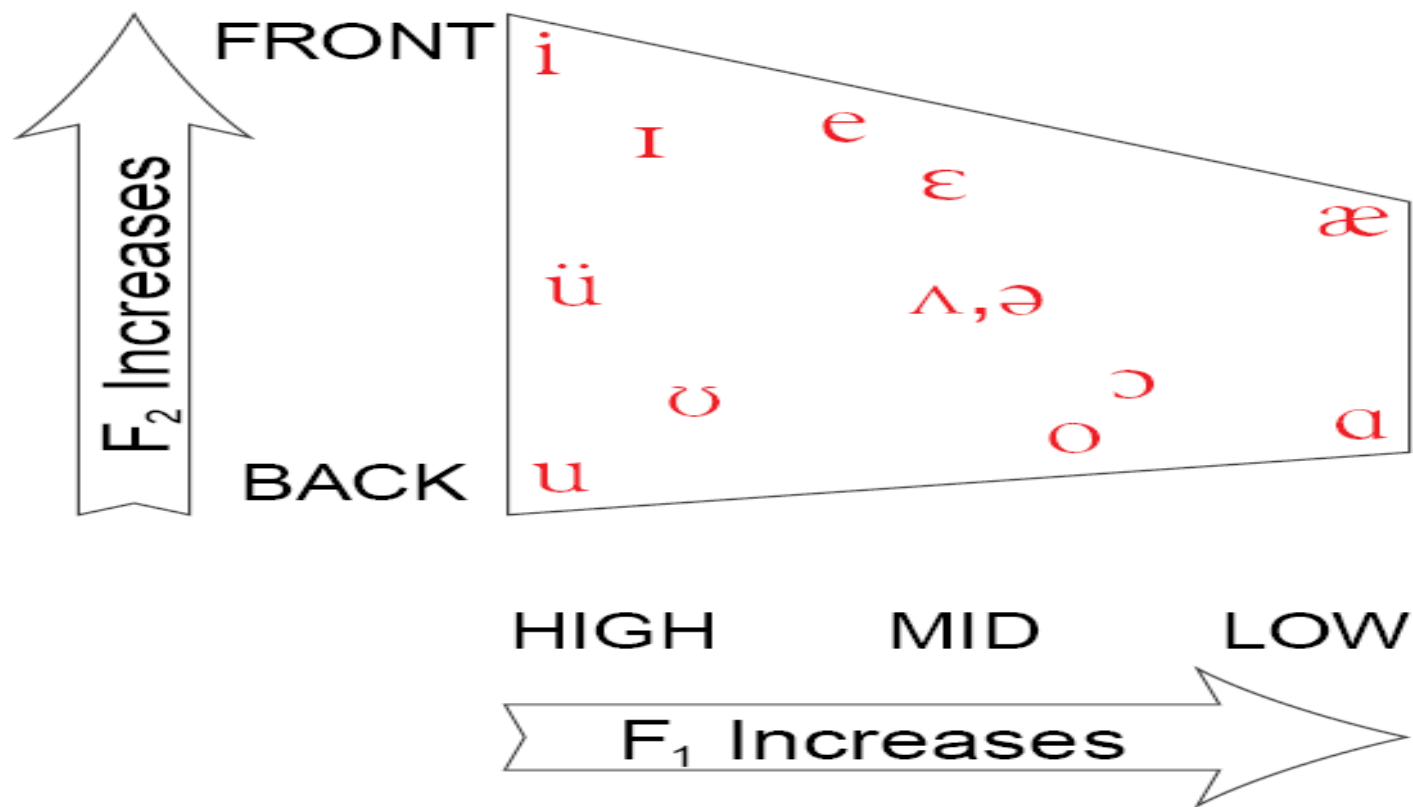
- Each vowel has a different intrinsic duration
- Schwa's have distinctly shorter durations (50ms)
- /ɪ, ɛ, ʌ, ʊ/ are the shortest monothongs
- Context can greatly influence vowel duration

Female Speakers

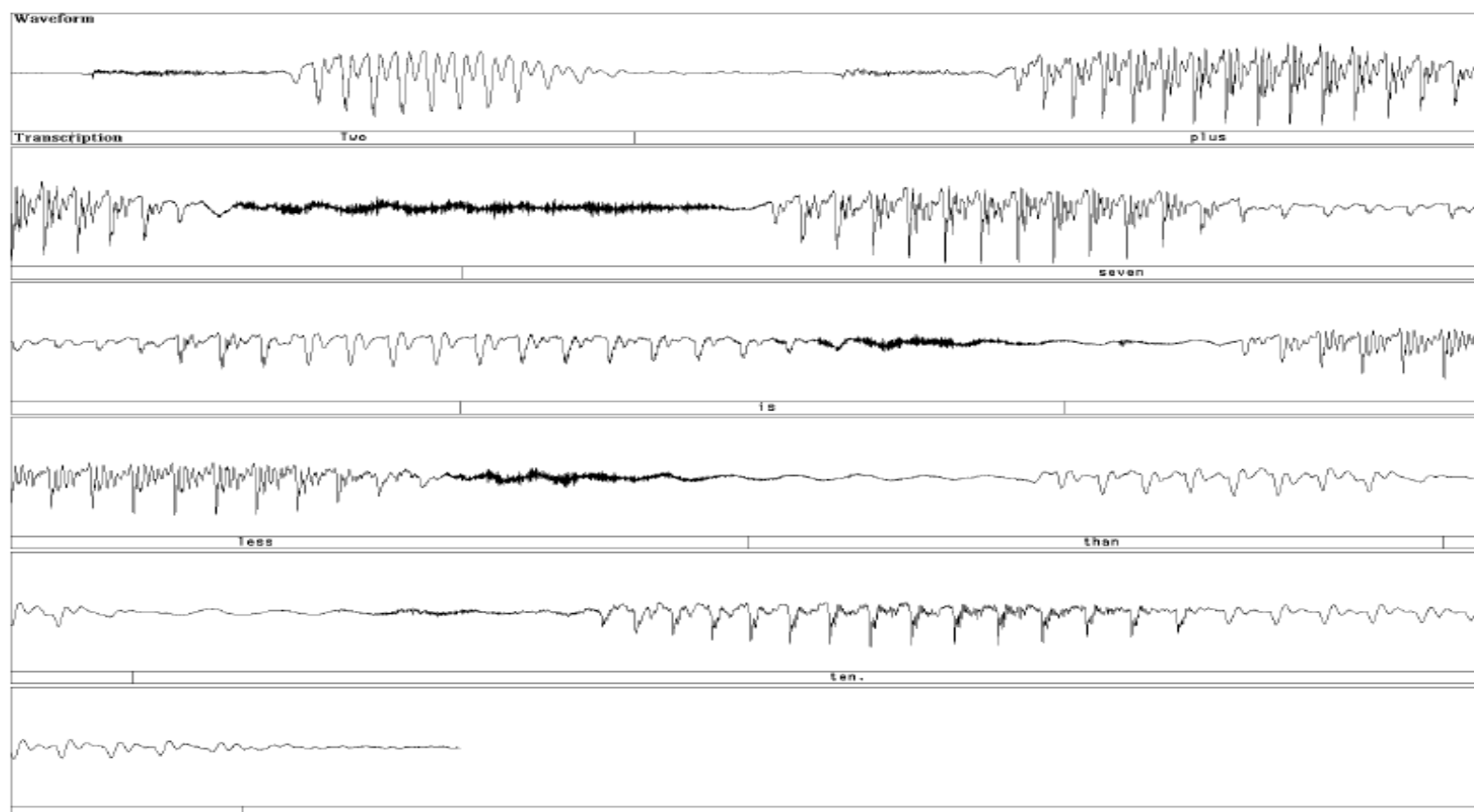


Male Speakers





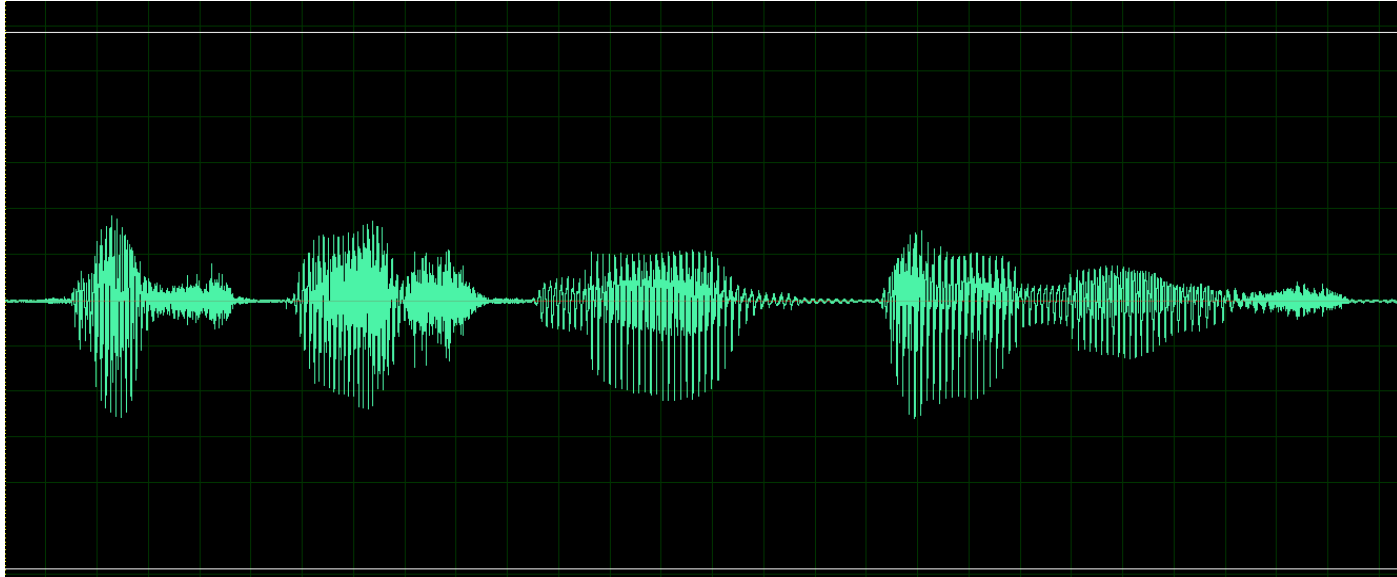
Speech Sample Waveform



Two plus seven is less than ten

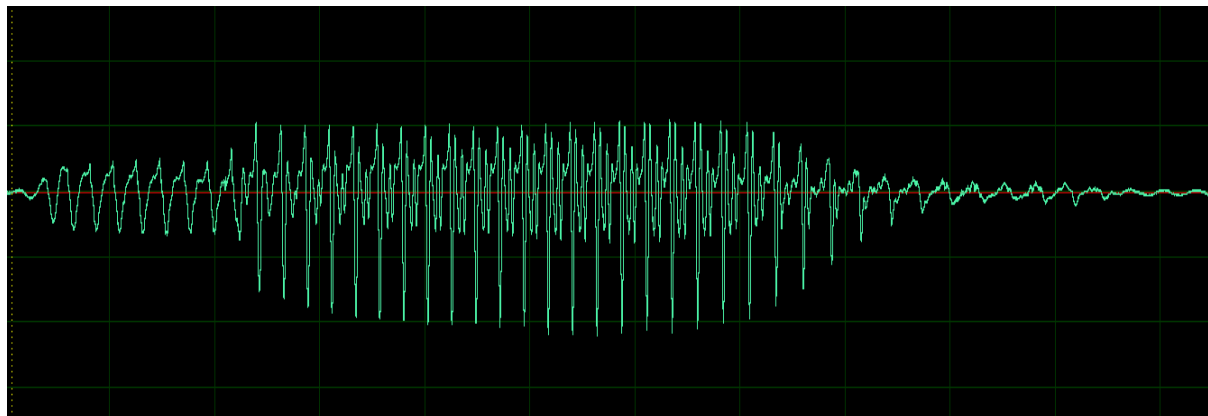
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شکل موج سیگنال گفتار — کلمات



اسمش ماهبانو است

شکل موج سیگنال گفتار-کلمات و واجها



ماه